

Chapter 2

LLM-Aided Large-Scale Data Annotation

Large-scale semantic annotation creates a major bottleneck between data collection and analysis. This chapter introduces Constrained Reasoning Instructions (CRIs), a structured prompting framework designed to guide large language models in performing feature-level semantic annotation. CRIs encode expert annotation diagnostics as modular instruction components that constrain model reasoning during inference. The system is first evaluated on a corpus of 347 English perfect-like grams (PLGs) annotated for 18 semantic features using 13 CRI configurations across three model families. Using the results from the preliminary experiments, a series of iterative refinements are implemented and tested resulting in a system that is shown to produce reliable annotations for features that depend primarily on surface cues and require minimal spot checking, while producing moderately reliable provisional annotations for more semantically nuanced features. These latter annotations require a more thorough vetting process before being implemented in subsequent experimental pipelines.

2.1 Introduction

Linguistic research increasingly relies on annotated corpora, but annotation often creates a bottleneck between data collection and analysis. High-quality annotation requires domain expertise, time, and sustained consistency, which is feasible on a small scale but introduces substantial labor and coordination costs when scaled up. This bottleneck is particularly acute for semantic annotation, where distinctions are analytically demanding, due to the gradience and context-sensitive nature of (the application of) semantic categories. In practice, semantic annotation requires annotators to operationalize theoretical distinctions on the fly, often resolving ambiguities left unspecified in the literature. As a result, semantic annotation demands both domain expertise and sustained cognitive effort. Prior work in computational semantics has long noted the difficulty of assigning context-sensitive meanings in annotation tasks (Artstein, 2017; Erk, 2012; Finlayson & Erjavec,

2017).

We illustrate this challenge through a case study in annotating temporal aspect and adjacent semantic categories. The study focuses on perfect-like constructions (or ‘grams’, following Bybee, 1986. Hence: PLGs), which have long posed difficulties for aspectual classification. Our broader project annotates PLG tokens in parallel corpora using an ontological feature modeling (OFM) framework that represents aspectual distinctions as bundles of semantic features. Each token is annotated for eighteen semantic features on the basis of a seed language (here: English), with interpretation depending on discourse context and interactions between features. At scale, this produces thousands of feature-level annotation decisions requiring careful contextual reasoning.

Recent work has begun exploring the use of large language models (LLMs) to assist linguistic annotation tasks, investigating how model inference can support or accelerate expert annotation workflows (Imamović et al., 2024). Other work has addressed the annotation bottleneck by developing improved annotation workflows and tooling designed to support expert annotators working with complex linguistic data (Rønningstad, Velldal, & Øvrelid, 2024). Building on these approaches, this chapter introduces Constrained Reasoning Instructions (CRIs), a structured framework which is being developed to assist in feature-level semantic annotation with LLMs. CRIs formalize linguistically grounded annotation diagnostics as structured instruction sets that guide model reasoning during inference. Rather than relying on unconstrained prompting, CRIs encode explicit decision criteria, edge cases, and illustrative examples that operationalize expert annotation procedures. CRIs are evaluated across multiple LLM families and permutations of the instruction components in order to analyze their effect on annotation performance. Additionally, we will see that formalizing annotation diagnostics in this way not only supports large-scale annotation workflows but also feeds back into the linguistic analysis, as it helps expose underspecified distinctions in the underlying semantic feature system.

The experiments presented here demonstrate that linguistically motivated diagnostic instructions can guide LLMs toward moderately reliable semantic annotation, while also revealing which types of diagnostic rules are most effective for model reasoning.

2.2 Related Work

Prior work in computational semantics has emphasized the importance of modeling meaning in context rather than assuming fixed lexical senses. For example, Erk (2012) propose contextualized distributional representations in which the semantic properties of a word are dynamically modulated by its surrounding context. This line of work highlights the central challenge of capturing context-sensitive meaning computationally. This dependence on contextual interpretation likewise complicates semantic annotation, where annotators must apply theoretical distinctions to individual instances whose meanings are shaped by discourse context.

Imamović et al. (2024) evaluate GPT-4 for fine-grained Appraisal annotation and report high precision but very low recall, weak sub-category agreement, instability across runs, and hallucinated

outputs. Their findings suggest that while LLMs can identify some evaluative items, unconstrained prompting yields inconsistent and incomplete pragmatic annotation. Similarly, Rønningstad, Vellidal, & Øvrelid (2024) evaluate GPT-4 for entity-level sentiment annotation in Norwegian and find moderate agreement with curated human labels, substantially below trained annotators. Performance improves with guidelines and few-shot examples, but the model requires post-processing and exhibits error patterns partially outside the range of human variation. Taken together, these findings suggest that while LLMs show promise for assisting annotation tasks, conventional prompting approaches often struggle to produce consistent and reliable semantic annotations.

Recent work has focused on improving annotation efficiency through workflow and tooling innovations. Rouzegar & Makrehchi (2024), for example, integrate LLMs into annotation workflows using active learning strategies that prioritize the most informative samples for human labeling, reducing the total volume of manual annotation required. At the same time, annotation tools such as EEEVEE (Sørensen, Peng, & Plank, 2024) emphasize usability, lightweight deployment, and token-level workflows aligned with NLP pipelines. These systems aim to reduce technical and cognitive overhead for annotators through simplified data formats and streamlined interfaces, improving the overall efficiency of human annotation environments.

While these approaches address important logistical challenges in large-scale annotation, they generally treat the underlying annotation reasoning as a given, meaning that the interpretive logic guiding label assignment is assumed to be specified in annotation guidelines rather than modeled or formalized by the system itself (Artstein, 2017; Finlayson & Erjavec, 2017). Their primary focus lies in optimizing workflow structure, interface design, and the allocation of human effort, essentially determining who annotates what and under what conditions. By contrast, the CRI framework operates at the level of interpretive decision-making itself. By formalizing expert diagnostic criteria as structured, machine-readable reasoning instructions, CRIs aim to constrain how annotation judgments are made across tokens and contexts, rather than optimizing the surrounding annotation workflow.

2.3 Overview of Approach

CRIs are structured, expertise-driven diagnostic instructions designed to elicit consistent annotation judgments from a generative language model. Rather than relying on open-ended prompting, i.e., where models are given general instructions without explicit criteria, decision rules, or structured intermediate steps, CRIs encode theoretically motivated distinctions into modular, inspectable components that guide model reasoning during inference in the same way they would guide human annotator reasoning. This structured design supports interpretability, systematic experimentation and iterative refinement.

2.3.1 CRI Components

Each semantic feature is associated with a CRI specification that follows a common structure, while the diagnostic content itself is tailored to the feature being annotated. Each CRI contains six components, each serving a distinct role in guiding the model’s reasoning.

First, the **intuition** introduces the feature descriptively, establishing in broad terms what the feature is intended to capture semantically. It is typically one or two sentences long and avoids explicit diagnostics, which are instead encoded in the following components.

Next, the **set_to_1_if** component establishing the positive diagnostics that indicate when a feature should be annotated as present. For example, a diagnostic from the Experiential feature reads: “The clause contains explicit frequency adverbs indicating repetition or distinct instances (e.g., ‘once’, ‘twice’, ‘often’, ‘ever’, ‘never’, ‘three times’).” If the target gram is modified by such a frequency adverbial, the model is instructed to assign a value of 1 for that feature.

Third, the **set_to_0_if** component specifies negative diagnostics indicating when a feature should be annotated as absent. For example, a diagnostic from the Unaccusative feature reads: “The target verb licenses or introduces an external argument (agent/initiator), whether overtly realized or suppressed.” If the target gram introduces an external argument, the model assigns a value of 0 for that feature. As discussed below, meta-instructions in the annotation pipeline guide the model in resolving cases where both positive and negative diagnostics appear to apply.

Fourth, because linguistic data often contain constructions that fall outside core diagnostics, the **edge_cases** component specifies how such instances should be handled. These rules are developed through iterative verification and refinement during annotation. In some cases the component may remain empty if no systematic edge cases have been identified for a feature.

Fifth and sixth, the **examples_positive** and **examples_negative** components provide canonical examples illustrating cases that should be annotated as present or absent. These examples directly reflect the diagnostics established in the preceding components and provide reference cases that help anchor the model’s interpretation of the feature.

2.3.2 Application of CRIs in Prompting

During inference, each annotation instance is evaluated using the target linguistic string including a perfective form, its surrounding discourse context, and the feature-specific CRI specification. These inputs are embedded within a prompt that includes task instructions and meta-instructions governing how the model should apply the diagnostics. The full CRI prompt template and meta-instructions used in the experiments are provided in the appendix.

The prompt first establishes the task context (e.g., “You are a trained linguist performing binary semantic feature annotation”), followed by instructions specifying that the model must assign a binary feature label based solely on the target gram and the surrounding discourse context. The prompt explicitly instructs the model to follow the CRI diagnostics as written and not to rely on external world knowledge or unstated assumptions.

Meta-instructions then specify how the diagnostics should be operationalized during reasoning. In the current implementation, the model is instructed to assign a value of 1 if at least one `set_to_1_if` condition applies. If a `set_to_0_if` condition applies, the model assigns 0, which overrides positive diagnostics unless an `edge_cases` rule explicitly specifies otherwise. These meta-instructions establish a consistent decision procedure across all features.

The model returns three outputs for each annotation decision: a binary feature label (0 or 1), a confidence rating (High, Medium, or Low), and a brief explanation summarizing the reasoning behind the classification. In this way, CRIs specify the reasoning procedure itself, while the prompt scaffold and meta-instructions enforce its execution during model inference.

2.4 Data & Methods

To evaluate the effectiveness of Constrained Reasoning Instructions for semantic annotation, a series of experiments was conducted applying CRIs to a corpus of perfect-like grams and measuring model outputs against human annotations and feature-level verification checks.

2.4.1 Data

The dataset used in this study consists of 347 tokens of perfect-like grams (PLGs) drawn from verses of the New Simplified Bible (2009), an English translation distributed through the Zefania XML Bible corpus (CrossWire Society). Bible data was used with an eye on cross-linguistic experimentation, given that the Bible has translations in over 1,300 languages (McCarthy et al., 2020). Each token corresponds to a single instance of an English perfect (*have*+participle) identified in the corpus. In some verses multiple tokens occur, but in each case the specific gram under evaluation is explicitly specified in the input provided to the model, ensuring that the system does not need to infer which element of the clause is the target of annotation.

2.4.2 Features for annotation

Each token was annotated for 18 semantic features derived from the linguistic literature on perfects and perfect-like grams (Bowler & Ozkan, 2017; Bybee & Dahl, 1989; Comrie, 1976; Iatridou, Anagnostopoulou, & Izvorski, 2001). The feature set operationalizes distinctions widely discussed in this literature, including canonical perfect readings (e.g., experiential, resultative), diagnostics related to argument structure, and Aktionsart-related properties that interact with perfect interpretation. Each feature is associated with a concise semantic definition used to guide annotation. Table 2.1 lists the features and their corresponding definitions¹, while the full CRI specifications used in the experiments are provided in the appendix.

¹ Two brief notes: (i) “diatam” is used here as a working label for perfect-like grams (PLGs), intended to reduce terminological ambiguity between *perfect* and *perfective* and to provide a more theory-neutral designation for this domain; (ii) “DTAM” refers to (past-referencing) definite temporal adverbial modification.

Feature	Definition
DtExp	Diatam: Experiential: The gram presents the event as part of the subject’s life history. It answers the question of whether the event has occurred at least once, without asserting that a resulting state currently holds.
DtRes	Diatam: Resultative: The gram indicates that a change of state has occurred and that the resulting state holds at the reference time. It encodes a transition from State A to State B, with the new state currently in effect.
DtUniv	Diatam: Universal: The gram expresses that a state, activity, or habitual series holds continuously from a past starting point up to the reference time.
TLTS	TAM Layering: Tense Shifted: This feature is present when the gram overtly encodes a tense or mood specification other than the simple present perfect. It distinguishes the canonical present perfect (‘have/has’ + past participle) from past, future, modal, or progressive perfect forms.
TLAspPfv	TAM Layering: Perfective Aspect: The gram presents the situation as a bounded whole, without making its internal temporal development accessible. If the predicate denotes a transition, the construal targets the transition itself as a single completed event.
TLMdIrr	TAM Layering: Irrealis Mood: The situation is construed as projected, hypothetical, imagined, desired, or otherwise non-actual. It is not asserted as a realized fact in the actual world.
AktAcc	Aktionsart: Accomplishment: A dynamic, durative event that unfolds over time and necessarily culminates in a defined terminal point or result state. The event includes an inherent process phase leading toward completion.
AktAch	Aktionsart: Achievement: A dynamic change-of-state event conceived as punctual or near-instantaneous. It represents the boundary point of transition itself rather than an unfolding process phase. The event is the crossing of a definable threshold.
AktAct	Aktionsart: Activity: A dynamic, durative situation that unfolds over time without an inherent endpoint or result state. It requires sustained dynamic engagement to continue, and interruption does not imply failure to reach a defined goal.
AktState	Aktionsart: State: A situation that holds over time without involving dynamic change or inherent progression. It has no inherent endpoint, and any subinterval of the situation is qualitatively identical to the whole.
ClNegPol	Clause: Negative Polarity: The gram presents the described event or state as not occurring or not holding. The situation is construed as absent from reality rather than realized.
ClCntfcl	Clause: Counterfactual: The clause describes a situation that is contrary to fact in the actual world. The event or state is understood not to have occurred, even though it is presented as possible under different conditions.
VcPass	Voice: Passive: The semantic patient of the event functions as the grammatical subject. The agent is either expressed in a by-phrase (e.g., ‘by John’) or omitted entirely. The construction includes overt morphological marking of the passive voice.
ASTrns	Arg. Struc.: Transitive: The lexical verb encodes a two-participant event involving an external argument (agent/initiator) and an internal argument (patient/theme), where the event is construed as directed from the external argument toward and affecting the internal argument.
ASUncc	Arg. Struc.: Unaccusative: The lexical verb encodes a one-participant event in which the sole argument functions as an internal argument (theme/patient), and the verb does not license an external argument.
ASUnrg	Arg. Struc.: Unergative: The lexical verb encodes a one-participant event in which the sole argument functions as the external argument (agent/initiator), and no internal argument is licensed.

Table 2.1: Semantic features and their definitions.

2.4.3 Experimental Design

To evaluate the role of individual components within the Constrained Reasoning Instruction framework, the system was tested on a 347-token dataset and using 13 unique combinations of CRI components. These configurations consisted of the full CRI specification as well as systematic perturbations of the six CRI components through both ablation experiments, in which a single component was removed, and ‘allation’ experiments, in which a single component was added. This design allows us to assess the contribution of each component to overall annotation performance. The experiments were conducted across three model families representing different LLM architectures: Anthropic (Claude-Haiku-4.5), Gemini (Gemini-2.5-Flash), and OpenAI (GPT-5-nano) models; all with a temperature setting of 0 (ensuring deterministic outputs), 512 max tokens, and no seed. In total, the study includes 702 experimental runs, corresponding to all combinations of the 18 features, 13 CRI configurations, and 3 model families. Each run consists of annotating the full 347-token dataset for a single feature under a specific CRI configuration and model family.

2.4.4 Evaluation

The token set was annotated for these semantic features by a human annotator. A randomly sampled subset ($N = 60$) of the tokens was annotated by a second human annotator. Both annotators were trained linguists. The evaluation metrics defined below, when applied to the two human annotators, help interpret the quality of the model outputs, as they provide an upper bound on how much we can expect the LLM-based annotations to agree with the human annotations.

Model outputs were evaluated using two complementary approaches. First, the model outputs were compared to ‘silver standard’ human annotations of the same dataset in order to measure how well the model matched the human judgments using standard evaluation metrics including precision, recall, F_1 score, Cohen’s κ , and Krippendorff’s α . In this context, precision reflects how frequently a model’s positive annotations agree with the human annotations (i.e., how many positives are true positives), recall reflects how frequently a model captures (or misses) true positives (i.e., how many false negatives does it produce), and the F_1 -score is the harmonic mean of precision and recall. Note that these metrics emphasize agreement on positive instances and may be sensitive to feature frequency and class imbalance, meaning that these scores should be interpreted as indicative rather than exhaustive measures of model performance.

To analyze the model outputs further, the step utilized feature verification checks designed to evaluate the internal coherence of the resulting feature assignments. Because the feature system encodes linguistically motivated constraints on how features can co-occur, certain feature combinations are typologically implausible. Model outputs were then examined for violations of these expectations.

Two types of verification checks were applied. The first examines patterns of co-occurrence and non-co-occurrence within ontological groupings such as perfect readings, argument structure properties, and Aktionsart classes. The second identifies typologically implausible feature pairings, such

		Claude					Gemini					OpenAI				
CRI Configuration		P	R	F_1	κ	α	P	R	F_1	κ	α	P	R	F_1	κ	α
Ablations	full CRI set	67.6	67.2	62.9	56.1	57.7	68.8	55.4	54.9	48.0	48.5	68.0	64.2	61.6	51.6	50.6
	intuition	68.5	68.6	63.5	53.7	52.7	66.1	55.9	55.9	48.0	49.1	69.5	63.9	62.2	52.4	51.5
	set_to_1_if	68.0	64.7	61.4	51.9	51.1	69.7	46.1	48.8	42.0	42.3	64.7	67.6	62.4	51.7	50.6
	set_to_0_if	66.9	69.4	63.9	56.4	57.9	57.5	75.0	61.6	48.3	45.7	64.5	68.2	62.9	52.3	51.5
	edge_cases	66.8	68.2	63.2	56.4	57.9	65.6	59.9	59.2	51.1	52.6	68.8	64.6	62.9	52.7	51.7
	examples_positive	68.1	65.9	61.5	54.6	56.0	66.0	55.4	55.6	45.0	43.2	65.8	62.4	59.8	49.8	48.5
	examples_negative	67.6	67.5	62.5	55.9	57.4	67.9	53.6	54.1	46.2	47.1	68.5	63.8	61.2	51.1	50.0
Allations	intuition	53.6	57.3	50.4	38.1	39.2	46.2	72.0	52.8	33.3	25.5	44.6	67.1	48.7	31.1	27.2
	set_to_1_if	56.6	72.7	59.8	47.8	48.7	56.0	66.6	57.8	42.4	39.6	55.7	70.3	58.4	44.6	42.9
	set_to_0_if	62.4	55.9	50.4	42.0	42.7	26.8	11.3	13.4	11.4	2.6	46.7	74.6	53.4	37.6	29.9
	edge_cases	48.5	64.2	50.5	34.3	28.7	44.2	45.5	38.4	20.6	12.1	39.6	69.5	44.7	23.8	6.5
	examples_positive	46.7	85.9	55.6	36.4	20.9	52.2	65.4	50.0	32.1	18.7	48.6	77.5	54.2	36.7	26.8
	examples_negative	54.6	39.7	38.9	27.6	22.3	30.7	12.2	14.5	9.0	-2.8	34.1	66.7	40.9	18.8	2.0

Table 2.2: LLM performance/LLM-annotator agreement broken down by **model** (column blocks) and **CRI configuration** (rows). P=Precision, R=Recall, F_1 = F_1 -score, κ =Cohen’s κ , α = Krippendorff’s α .

as combinations of perfective aspect with incompatible state/activity/universal readings. These checks provide a complementary evaluation perspective by assessing whether the model’s predictions produce linguistically coherent feature bundles, even when they diverge from the human annotation.

2.5 Preliminary Experiment & Results

2.5.1 Evaluation for CRI component configurations

The central question of this phase of the project is whether the six-component framework of CRIs provides an approach to prompt-based annotation that leads to high-quality model output. Table 2.2 summarizes model performance across the thirteen CRI configurations. When looking at the performance and agreement when using the full CRI set, Claude achieves the highest average F_1 across permutations at 62.9, followed closely by OpenAI (61.6), while Gemini performs somewhat lower overall (54.9), with chance-corrected agreement measures (Cohen’s κ and Krippendorff’s α) following a similar pattern, both having values for Claude and OpenAI that are typically interpreted as indicated moderate-to-substantial agreement. Model–human agreement approaches the level of human–human agreement at an F_1 -score of 64.0, compared to 62.9 (Claude), 54.9 (Gemini), and 61.6 (OpenAI) for the full-CRI models.

A similar pattern is observed across metrics and feature-level results, with model performance

typically falling a few points below human agreement, indicating that while CRI-guided annotation is competitive, it does not fully match human consistency.

When results are examined by CRI permutation, a notable result is that the removal of the `set_to_0_if` component consistently yields the highest macro-averaged F_1 scores across the three systems (63.9 for Claude, 61.6 for Gemini, and 62.9 for OpenAI). This suggests that the ‘negative rules’ may introduce interactions that complicate model reasoning rather than consistently improving it, especially for Gemini, where the difference in F_1 score between the full CRI set and the full set minus the `set_to_zero_if` component is close to 7 points. This effect is especially noteworthy given the intended role of these negative rules as veto conditions designed to prevent over-assignment of features.

Looking at CRI component configurations retaining only a single CRI component (the ‘allation’ conditions), it was found that the individual components are able to shoulder different degrees of agreement with the human annotations by themselves. Most CRI components by themselves lead to a steep drop in performance, of 10 points or more from the best-performing configuration per LLM. Notably, only specifying the positive rules by themselves (`set_to_1_if`) does not incur the same drop in performance, suggesting that this is a particularly informative component of the CRI set. However, this component by itself never leads to the best performance, suggesting that the other components contribute over and above it.

2.5.2 Evaluation for individual features

Having evaluated the model output in the aggregate, the next step is to examine the individual semantic features, using the full CRI configuration. Table 2.3 shows the results, broken down per feature and per model. These results reveal substantial variation in model performance across the feature inventory. Several features show consistently high agreement across models, including ASTrns (transitives), ClCntfcl (counterfactuals), ClNegPol (negative polarity), TLAspPfv (perfective aspect), and VcPass (passive voice), with F_1 scores frequently exceeding 85 for at least one model and agreement coefficients approaching or exceeding 80 in the strongest cases. These features correspond to distinctions with relatively clear grammatical or lexical cues in the input text.

Other features prove considerably more difficult for the models. In particular, several Aktionsart values, including accomplishments, activities, and states, as well as DtRes (resultatives), exhibit unstable precision–recall trade-offs across models, sometimes producing high precision but very low recall, or the reverse. Interestingly, achievements show somewhat stronger performance than the other Aktionsart categories despite their close semantic relationship to accomplishments.

Perhaps the most consistently challenging feature across models is ASUnrg (unergatives), which yields low F_1 scores across systems. The contrast between the strong performance of ASTrns (transitives) and the weaker performance of ASUnrg suggests that the difficulty may lie less in the underlying linguistic distinction than in how the relevant diagnostics are operationalized within the CRI specification. Distinguishing these categories appears to require finer semantic interpretation that is less reliably recoverable from surface cues alone.

Feature	Claude					Gemini					OpenAI				
	P	R	F_1	κ	α	P	R	F_1	κ	α	P	R	F_1	κ	α
ASTrns	90.2	94.7	92.4	72.9	72.9	95.9	76.6	85.2	60.6	59.8	95.7	81.9	88.3	66.6	66.3
ASUncc	91.4	74.4	82.1	76.9	76.9	95.7	52.3	67.7	60.8	60.1	86.4	81.4	83.8	78.7	78.7
ASUnrg	33.3	41.2	36.8	33.2	33.3	28.6	47.1	35.6	31.4	31.2	23.1	17.7	20.0	16.5	16.5
AktAcc	51.4	44.6	47.7	32.8	32.8	64.3	10.8	18.6	12.5	5.5	63.6	33.7	44.1	33.0	31.7
AktAch	74.0	93.1	82.5	56.1	55.3	78.7	45.5	57.6	29.9	26.3	73.2	75.9	74.5	43.6	43.6
AktAct	83.3	27.8	41.7	38.5	37.4	61.5	22.2	32.7	28.7	27.6	100.0	19.4	32.6	30.2	28.2
AktState	84.6	25.6	39.3	35.6	34.1	75.0	7.0	12.8	10.9	6.5	60.5	53.5	56.8	51.1	51.2
ClCntfcl	77.1	96.4	85.7	84.3	84.3	84.4	96.4	90.0	89.1	89.1	90.0	96.4	93.1	92.5	92.5
ClNegPol	82.4	97.7	89.4	87.7	87.7	92.7	88.4	90.5	89.2	89.2	76.9	93.0	84.2	81.7	81.7
DtDTAM	5.8	80.0	10.8	8.4	0.3	5.7	60.0	10.3	7.9	2.3	6.3	80.0	11.6	9.2	2.0
DtExp	70.4	63.3	66.7	50.4	50.4	69.7	70.8	70.3	54.3	54.4	68.5	74.2	71.2	55.0	55.1
DtRes	86.4	70.8	77.8	60.1	59.9	100.0	19.9	33.2	20.1	5.3	90.8	63.2	74.5	57.2	56.2
DtUniv	75.9	38.6	51.2	45.1	44.3	65.0	45.6	53.6	46.3	46.2	76.7	40.4	52.9	46.9	46.2
TLAspPfv	96.0	82.1	88.5	49.8	48.8	97.2	72.4	83.0	40.7	37.4	97.3	75.2	84.8	44.1	41.6
TLMdIrr	76.5	88.6	82.1	79.3	79.3	81.3	88.6	84.8	82.5	82.5	70.6	81.8	75.8	72.0	72.0
TLTS	50.3	98.7	66.7	53.6	51.4	57.5	98.7	72.6	62.5	61.5	58.9	98.7	73.7	64.2	63.3
VcPass	87.8	92.3	90.0	88.7	88.7	86.1	94.9	90.2	88.9	89.0	85.4	89.7	87.5	85.9	85.9

Table 2.3: LLM performance/LLM-annotator agreement broken down by **model** (column blocks) and **feature** (rows). P=Precision, R=Recall, F_1 = F_1 -score, κ =Cohen’s κ , α = Krippendorf’s α .

Across features, models often exhibit somewhat higher recall than precision, suggesting a tendency in some cases to predict feature presence more readily than absence. At the same time, the improved performance observed when `set_to_0_if` rules are removed indicates that the current formulation of negative veto conditions may introduce additional reasoning difficulty for the models.

One possible explanation for the performance gains observed when ablating the `set_to_0_if` component lies in the interaction between feature-specific diagnostics and the global meta-instructions embedded in the prompting scaffold. In the current pipeline, the prompt includes a general decision rule stating that negative diagnostics veto positive ones unless overridden by an edge-case instruction. Because the `set_to_0_if` block already encodes feature-specific negative diagnostics, the system effectively introduces two layers of negative constraint: explicit feature-level veto conditions and global reasoning rules governing how those conditions should be interpreted. For a generative language model performing stepwise reasoning, this redundancy may increase instruction complexity rather than strengthening the constraint. When the `set_to_0_if` component is removed, the model can rely primarily on positive diagnostics together with the global decision rules, potentially simplifying the reasoning process and reducing instruction interference. This pattern suggests that the observed improvement may stem from redundancy between the CRI diagnostics and the global veto rule, raising the possibility that future implementations could instead simplify the prompt-

ing scaffold by removing or revising the meta-instruction layer while retaining the feature-specific diagnostic structure.

The feature-level results further demonstrate that model performance aligns with the interpretive demands of the underlying diagnostics: features relying on clearer structural cues show stronger agreement, while those requiring more subtle semantic inference remain challenging.

2.5.3 Feature Verifications

In addition to aggregate evaluation metrics, the annotation pipeline includes a verification layer designed to support rapid inspection of individual token–feature assignments. For several feature groups, such as perfect readings, argument structure, and Aktionsart, each token is expected to receive exactly one positive value. This allows the system to automatically identify both over-assignment (multiple positive values within a cluster) and cases where no value is assigned (“hot potatoes”). Additional checks flag marked or typologically unusual pairings, such as combinations of perfective aspect with features that rarely co-occur, such as states and universal perfect readings. These verification mechanisms provide a lightweight but effective way to detect annotation anomalies and inspect model behavior at the token level, complementing the aggregate performance metrics reported above. A full table of verification test results can be found in the appendix.

Several consistent patterns emerge from the verification results. First, in the full CRI condition, violation rates are generally low across most verification categories, with mutually exclusive feature assignments (e.g., multiple perfect readings or conflicting argument structure labels) typically remaining in the low single digits for Claude and OpenAI, while Gemini exhibits consistently higher rates in the same condition. This pattern indicates that the full CRI specification can effectively enforce basic structural constraints on model output, though its effectiveness varies across models.

Second, violation rates remain comparatively low under ablation conditions, and in some cases are comparable to or lower than those observed in the full CRI configuration. By contrast, configurations retaining only a single CRI component (the allation conditions) show a sharp increase in violations. In these settings, multiple-assignment errors (e.g., AktMultiple) and constraint violations in argument structure and perfect readings frequently rise to very high levels, in some cases affecting a majority of tokens. This pattern suggests that while individual components may contribute useful diagnostic signals, no single component is sufficient to maintain constraint satisfaction on its own, and that stable model behavior instead emerges from the interaction of components, which function together synergistically to enforce the intended constraints.

Third, when focusing on the higher-performing configurations (the full CRI condition and ablations), constraint satisfaction is consistently strong across several feature domains. Violations involving mutually exclusive perfect readings remain generally low for Claude and OpenAI, with Gemini again showing substantially higher rates under the same conditions. Argument structure constraints are also handled robustly, with very low rates of conflicting assignments across models, and typologically marked perfective pairings (“red flag” combinations) are likewise rarely produced.

Together, these results indicate that the CRI framework is broadly effective at enforcing both structural exclusivity and cross-feature compatibility constraints, particularly when multiple components are present.

A different pattern emerges, however, when considering omission errors. Across models and configurations, “hot potato” cases (no value assigned) are consistently more frequent than overlapping assignments within a feature cluster. This asymmetry suggests that models are more likely to withhold an assignment than to risk violating a constraint, particularly in the presence of competing or redundant signals. This tendency aligns with the earlier observation that doubling the layers of negative constraints may introduce interaction effects within the prompting scaffold, increasing uncertainty and biasing the model toward under-assignment rather than over-assignment.

A final caveat concerns the nature of the verification procedure itself. Because these checks evaluate internal consistency within model outputs, rather than accuracy against an external gold standard, they cannot detect cases where multiple related assignments are jointly incorrect (e.g., an erroneous positive assignment paired with an equally erroneous omission elsewhere in the same feature cluster). While such cases are plausibly less frequent than single-sided errors, their true prevalence remains an open question. For this reason, the verification layer is best understood as a diagnostic tool rather than a substitute for evaluation: its primary value lies in enabling targeted manual inspection, through which recurring error patterns can be identified and used to iteratively refine both the CRIs and the broader annotation framework.

These verification results show that, in this setup, the system is able to enforce structural constraints and cross-feature compatibility to a substantial degree, while also revealing systematic patterns in model failure, especially in the form of omission errors under constraint interaction. At the same time, performance remains sensitive to both the feature being annotated and the specific configuration of the instruction set.

2.5.4 Summary of Preliminary Results

Two main observations emerge from these results. First, the experiments demonstrate that structured diagnostic prompting can guide large language models toward linguistically plausible semantic annotations, even when the target categories involve relatively subtle distinctions in aspect and event structure. Several features achieved high agreement with expert annotations, suggesting that LLMs can successfully apply certain types of linguistic diagnostics when these are explicitly encoded in the prompt.

Second, the CRI permutation experiments provide insight into how models interpret diagnostic instructions. Contrary to expectations, removing the negative constraint rules (`set_to_0_if`) improved performance across all models. One possible explanation is that the current prompting pipeline introduces redundant constraint layers: feature-specific negative diagnostics within the CRIs and global veto rules embedded in the prompt scaffold. For a model performing stepwise reasoning over instructions, this duplication may increase decision complexity rather than making the constraint easier for the model to operationalize. When the `set_to_0_if` component is removed,

the model can rely more directly on positive diagnostics together with the global decision logic, potentially simplifying the reasoning procedure and yielding more stable behavior. This result highlights the importance of carefully balancing rule complexity and prompt specificity when designing structured reasoning instructions.

Taken together, these findings show that CRIs are best understood not as a static solution, but as an iterative framework whose effectiveness depends on careful design, empirical validation, and ongoing refinement within a given annotation context. Thus CRIs function most effectively as a diagnostic scaffolding that directs model attention toward relevant contextual cues rather than as a fully specified rule system. In practice, this supports a workflow in which the model generates provisional annotations that can be rapidly verified and corrected by a human annotator.

2.6 System Refinements

In order to implement the system in the larger project pipeline, i.e., assisting in data preparation for the OFM phase, it was necessary to examine its behavior in more detail in order to determine not only how to improve its performance, but also to better understand how it can be used most effectively. This section describes the process by which detailed error analysis of the preliminary results was used to inform refinements to the system, and then presents the results of subsequent tests on the refined system in order to set the stage for the final section of this chapter in which the implementation strategy is presented.

The first step in this process was to conduct token-level error analysis in order to identify patterns in the data that can inform how the CRIs are crafted. First, the two top-performing models, Anthropic and OpenAI, were run again using the same CRIs but removing the meta-instructions to determine the effects they had on the experiments, specifically with respect to the hypothesis that interaction between meta-instructions and `set_to_0_if` rules contributed to deleterious effects in the models’ reasoning (see §2.5.2). These experiments were conducted only on the full and ablation permutations of the CRIs, as the ablation tests were reliably poor performers.

The results of the round 2 (“R2”) experiments did show some improvement over the originals (“R1”), but only modestly so, as seen in Table 2.4. Note that the total number of annotations conducted by each model/permutation configuration is 6,246 (18 features $\times$ 347 tokens). As omission of the meta-instructions yielded at best a 10% improvement on the original results, it is unlikely that they are a significant contributing factor to overall error (i.e., disagreement) rates. The next phase of refinement involved token-level surveys of the results for each feature and model; it was determined that subsequent experiments would focus on the full set of CRIs and the Anthropic model, given the overall strength of that configuration’s performance.

2.6.1 Token-level survey

Errors identified in the token-level survey fell into three broad categories: (i) CRI formulation issues, where model behavior reflected ambiguities or unintended generalizations in the instruction

Model_Permutation	Round 1		Round 2		Comparison	
	Total Dis.	%of Total	Total Dis.	% of total	Total Dis.	% of total
Anth_full	736	11.8%	675	10.8%	-61	+1.0%
Anth_abl1	716	11.5%	678	10.9%	-38	+0.6%
Anth_abl2	781	12.5%	720	11.5%	-61	+1.0%
Anth_abl3	732	11.7%	662	10.6%	-70	+1.1%
Anth_abl4	765	12.3%	731	11.7%	-34	+0.5%
Anth_abl5	753	12.1%	685	11.0%	-68	+1.1%
Anth_abl6	739	11.8%	705	11.3%	-34	+0.5%
OpenAI_full	763	12.2%	805	12.9%	+42	-0.7%
OpenAI_abl1	808	13.0%	842	13.5%	+34	-0.5%
OpenAI_abl2	779	12.5%	771	12.3%	-8	+0.1%
OpenAI_abl3	777	12.4%	745	12.0%	-32	+0.5%
OpenAI_abl4	796	12.7%	751	12.0%	-45	+0.7%
OpenAI_abl5	787	12.6%	781	12.5%	-6	+0.1%
OpenAI_abl6	777	12.4%	744	12.0%	-33	+0.5%

Table 2.4: Performance comparison of Round 1 versus Round 2 (omitting meta-instructions), total disagreements and percentage of total annotations.

wording; (ii) diagnostic interpretation issues, where models applied underspecified semantic concepts too broadly; and (iii) text-level interpretation errors, where disagreement arose from the model’s interpretation of the input rather than from misapplication of the CRIs. Each error type is both methodologically and linguistically informative, and motivates different approaches to refinement.

Some model reasoning errors were attributable to factors in the framing and wording of each feature’s CRIs, which could be addressed using minor tweaks. For example, the DtDTAM feature was originally worded in such a way as to capture clausal temporal adverbials, such as “when the party ended, we had already left”, in which *when the party ended* serves as a temporal anchor whose inclusion was justified by way of the observation that such adverbials trigger Present Perfect Puzzle (Klein, 1992) effects: */# “when the party ends/ended, we have already left”. However, a survey of the results showed that wording of the CRIs induced the model to determine that any clausal temporal reference anywhere in the broader context was sufficient to annotate a token as expressing DTAM. This phenomenon is exemplified in (1).

- (1) Overgeneralization of clausal adverbial rule:

Feature: DtDTAM (Definite Temporal Adverbial Modification)

CRI wording: “The clause contains a clausal past time anchor starting with temporal conjunctions (e.g., ‘when’, ‘after’, ‘before’, ‘since’ [temporal]).”

Example text: “When you abstain from food, do not be like the hypocrites, with sad faces to be seen by men. They have received their reward.”

Model reasoning: “The clause contains a clausal past time anchor ‘When you abstain from food’ which establishes a specific temporal relationship for the event ‘have received’, satisfying the requirement for an overt temporal adverbial.”

In addition to the rather broad interpretation of where the clause begins and ends, the model’s reasoning suggests that the framing itself is too broad, i.e., the presence of any clausal temporal anchor in the “vicinity” of the target gram expresses DTAM. This was addressed by tightening the framing to say, “The target gram is directly modified by a clausal past time anchor...”. This adjustment significantly reduced the number of errors of this kind, though ultimately unrelated complications with this feature and the overall very low frequency of this feature’s occurrence motivated removing the DtDTAM feature from this phase of the project².

A similar issue arose with the TLTS (tense shifted) feature wherein the CRI text “The gram includes ‘been’ followed by a progressive participle” triggered the model to interpret all occurrences of *be* + (any kind of) participle as tense shifted. Because the New Testament text used in these experiments makes extensive use of passive constructions (i.e., *be* + past participle), the model’s determination that all cases of passives counted as tense shifted, even when the auxiliary was in the simple present tense, led to a significant number of annotation errors for this feature. This was remedied by including the specification that only participles in the *-ing* form count for the purposes of this feature.

Somewhat more challenging to address were reasoning errors arising from nuances of interpretation, particularly with respect to concepts like “durative process phase” (relevant to Aktionsart features) and “result state” (relevant to PLG reading features as well as to Aktionsart features). For example, the model tended to grossly over-interpret instances of PLGs as instantiating result states when they were not present, instantiating what I refer to as the “state of having verb-ed” problem. For example, the model might interpret the gram “I have spoken for an hour” as introducing the “result state” of having spoken for an hour. While a philosophical argument can be made that such a result state exists, such a broad interpretation of the term renders it virtually meaningless to linguistic analysis. An example of this type is shown in (2).

- (2) Overly broad interpretation of “result state”:

Feature: DtRes (Resultative)

CRI wording: “The verb describes a change of state (e.g., ‘break’, ‘die’, ‘melt’, ‘heal’, ‘wither’), and the undergoer is currently in that changed state.”

² The DtDTAM may be reintroduced at a later stage, should results of the OFM experiments motivate doing so.

Example text: “This has taken place so that the writings of the prophets will come true. Then all his disciples left him.”

Model reasoning: “The verb ‘take place’ describes an event that has occurred and the resulting state (that the event has happened) persists at the reference time, satisfying the resultative feature.”

These types of errors reflect the depth of nuance that many linguistic concepts bear, reinforcing the importance of careful and explicit definition of terms. In attempting to craft a truly unambiguous, machine-readable definition of a concept like “result state”, we are confronted with the role that human intuition and, for lack of a better term, “vibes” play in the interpretation of linguistic meaning. It is beyond the scope of this dissertation to provide a philosophically rigorous definition of what precisely entails a “result state”, or to claim that the current world is in the result of state of its past having happened, but in order to conduct meaningful linguistic analysis some sort of boundary lines must be drawn in order to capture the domain in question. In this particular case, the relevant CRI was update to include the text, “...and the resulting state is a distinct state introduced by the predicate (not merely the fact that the event has occurred).” When the token exemplified in (2) was re-run, the model returned the much more diagnostically useful reasoning given in (3).

- (3) Revised model reasoning: “The verb ‘take place’ is an activity/event verb that does not introduce a result state persisting at the reference time; it merely reports that an event occurred in the past without encoding a transition to a new state currently in effect.”

Issues of this type were addressed in the final refinement phase by careful, targeted tweaks to the wording that make these concepts more explicit, though some interpretive errors persisted. (See §2.6.4 for details.) One possible solution to this type of issue would be to construct a small, targeted “glossary” of particularly common and easily misconstrued terminology, such as “result state” and “durative process phase”. It is possible, however, that the introduction of another layer of complexity to the system may generate unforeseen consequences, and so such a glossary has not been developed at this stage, but the concept remains noted as a possible avenue for future system development.

The third and most challenging type of issue to address was similar to the previous one, but rather than the model misreading CRI instructions that can be clarified, errors instead arose from the model’s misinterpretation of the text itself. For example, in the case of the DtExp feature, some errors arose from misinterpreting the focus of the discourse being on the event versus the result state of the event. This is exemplified in (4).

- (4) Misinterpretation of linguistic context:

Feature: DtExp (Experiential)

CRI wording: “The verb denotes perception or communication where the focus is on the occurrence of an experience at least once, rather than on a resulting state or transfer.”

Example text: “For there are men who, from birth, were without sex. There are some who were made that way by men. There are others who have made themselves so for the kingdom of heaven. He who is able to accept this let him accept it.”

Model reasoning: “The verb ‘have made themselves’ denotes a bounded activity (self-castration or renunciation of sexuality) where the focus is on having undergone this experience as part of the subject’s life history for a religious purpose, rather than on a persisting physical result state.”

Such cases present challenges as the issue arises from the model’s ability to interpret the text more so than it’s ability to correctly apply the CRI diagnostics. Iterative testing showed that increasingly specified or explicit framing in the CRIs was able to improve performance in this area, but only to a limited degree, and that beyond a certain point too much specificity can in fact trigger confusion in the reasoning ultimately leading to overall degraded performance. For the purposes of system implementation in the larger project pipeline, it was determined that different degrees of human supervision are required for evaluating LLM annotations: some features, primarily those that involve analysis of surface forms (syntactic structure, morphological marking, etc.) can be relied upon to yield overall strong results and can be evaluated quickly and at scale using the verification system development for this purpose, while other, more nuanced features require a higher degree of strategic token-level evaluation. (See §2.7 for details.)

While errors which arose from the wording and content of the CRIs were generally easy to address by working backwards from the model’s stated reasoning, it should be noted that some researchers have documented examples of models providing reasoning which does not reflect the actual decision process, but rather serves as more of a post hoc justification for those decisions (CITE). Thus far in the course of this project the model’s reasoning that has been manually checked not only directly reflects the CRI framing but has also shown to reliably represent the model’s reasoning process as seen in demonstrated reasoning improvements (cf. (3)). Errors arising from conceptual ambiguity, as in the case of result states, show partial responsiveness to CRI refinement, though with diminishing returns suggesting that there is a sort of “Goldilocks zone” in which CRI specification is maximally instructive without introducing confusion into the model’s reasoning. In this way, the process of iterative refinement balances the pressures of addressing CRI specification, data ambiguity, and model reasoning tendencies.

Lastly, those errors which arise from textual misinterpretation are at this stage subject to limited refinement, and thus more fine-grained human oversight is required to verify that annotations are correct. The verification system (see §2.5.3) established for this methodology serves to “red flag” instances that are likely cases of erroneous reasoning, though for now detailed human supervision remains necessary.

2.6.2 Final results

Once error analysis had been completed, and the CRIs had been refined and individually tested, a final set of experiments was run using Anthropic (as the consistently best performing model) and the full set of CRI components. As intermediate stages of targeted testing revealed that omission of the meta-instructions led to slight but noticeable degradation of overall accuracy, all subsequent experiments included the meta-instructions in the prompt. In addition to adjustments made to the CRIs themselves, one additional system-level modification was made: the `edge_cases` component, having been shown to have deleterious effects on reasoning as shown by consistently strong performance of the ablation of that component, was reconfigured as an “exceptions” component, reserved exclusively for cases which directly contradict the stated `set_to_1` and `set_to_0` rules; all other diagnostics previously housed in this section by virtue of being rare were moved to the respective `set_to_1` and `set_to_0` sections.

Table 2.5 shows the results of the refinements as compared to the initial tests. Note that the `AktSemel` and `DtDTAM` features were omitted due to very low frequency of occurrence³. Overall, significant improvements were made across features in the refinement process, though a number of features are clear exceptions, at least with respect to F_1 score; see §2.6.3 for more. The greatest improvement was seen in the `AktState`, `AktAcc`, and `TLTS` features, at +32.9, +12.1, and +9.7 to F_1 respectively.

Features whose performance rose significantly (including `AktState`, `AktAcc`, and `TLTS`) were achieved primarily through refinement of CRIs where the original wording was overly broad, potentially ambiguous, or too narrow. For example, the `AktState` feature benefited from the addition of a `set_to_1_if` rule targeting durative attributes (e.g., ‘be tall’, ‘be a good person’), as well as an adjustment to a `set_to_0_if` rule that was designed to prevent progressive-marked grams from being annotated as states that had the unintended consequence of preventing *all* occurrences of *be* + participle from being included. These factors together account for the 45-point recall improvement seen between the preliminary and final experiments. A similar issue with the `TLTS` feature, as described in the previous section, triggered the erroneous marking of simple present passives as `TLTS`; an update to tighten the framing of the relevant CRI led to a 13.1-point improvement to that feature’s precision score.

Features whose performance showed notable apparent degradation between preliminary and final testing, such as `ASTrns`, `AktAct`, and `DtUniv` reflect difficulties in balancing over- and under-generalization (as with `ASTrns` and `DtUniv`) and in capturing subtleties of nuance (as with `AktAct`). In the case of `ASTrns`, a recurrent issue in the preliminary experiments was the model’s tendency to “hallucinate” the presence of a direct object or passive morphology when none was present. Minor adjustments that were made to tighten the constraints on such phenomena succeeded, as shown by the 3.1-point improvement to precision, but had the concomitant effect of reducing the recall score by 9 points. Note that while some features did see reduction in their F_1 scores, the overall degree of improvement far exceeds the scope of losses in accuracy.

³ There were no instances of `AktSemel` in the 347-token set, and only one instance of `DtDTAM`.

Feature	Anthropic, Preliminary					Anthropic, Refined					Comparison				
	P	R	F_1	κ	α	P	R	F_1	κ	α	P	R	F_1	κ	α
ASTrns	90.2	94.7	92.4	72.9	72.9	93.3	85.7	89.3	67.1	67.0	+3.1	-9.0	-3.1	-5.7	-5.8
ASUncc	91.4	74.4	82.1	76.9	76.9	92.0	80.2	85.7	81.4	81.4	+0.6	+5.8	+3.7	+4.5	+4.5
ASUnrg	33.3	41.2	36.8	33.2	33.3	42.9	35.3	38.7	35.9	35.9	+9.5	-5.9	+1.9	+2.6	+2.7
AktAcc	51.4	44.6	47.7	32.8	32.8	50.4	73.5	59.8	43.9	43.2	-1.0	+28.9	+12.1	+11.1	+10.3
AktAch	74.0	93.1	82.5	56.1	55.3	73.6	89.3	80.7	52.8	52.2	+0.5	-3.7	-1.8	-3.4	-3.1
AktAct	83.3	27.8	41.7	38.5	37.4	69.2	25.0	36.7	33.0	32.0	-14.1	-2.8	-4.9	-5.4	-5.4
AktState	84.6	25.6	39.3	35.6	34.1	89.7	60.5	72.2	69.1	69.1	+5.0	+34.9	+32.9	+33.6	+35.0
ClCntfcl	77.1	96.4	85.7	84.3	84.3	84.4	96.4	90.0	89.1	89.1	+7.2	-	+4.3	+4.7	+4.8
ClNegPol	82.4	97.7	89.4	87.7	87.7	82.4	97.7	89.4	87.7	87.7	-	-	-	-	-
DtExp	70.4	63.3	66.7	50.4	50.4	70.5	71.7	71.7	55.6	55.7	+0.1	+8.3	+4.4	+5.2	+5.2
DtRes	86.4	70.8	77.8	60.1	59.9	86.8	65.5	74.7	56.0	55.4	+0.4	-5.3	-3.1	-4.1	-4.4
DtUniv	75.9	38.6	51.2	45.1	44.3	79.2	33.3	46.9	41.2	40.0	+3.3	-5.3	-4.2	-3.9	-4.2
TLAspPfv	96.0	82.1	88.5	49.8	48.8	95.9	81.0	87.9	48.2	47.0	-0.1	-1.0	-0.6	-1.6	-1.8
TLMdIrr	76.5	88.6	82.1	79.3	79.3	80.0	90.9	85.1	82.8	82.8	+3.5	+2.3	+3.0	+3.5	+3.5
TLTS	50.3	98.7	66.7	53.6	51.4	63.4	95.9	76.3	68.1	67.7	+13.1	-2.7	+9.7	+14.6	+16.3
VcPass	87.8	92.3	90.0	88.7	88.7	83.7	92.3	87.8	86.2	86.2	-4.1	-	-2.2	-2.5	-2.5

Table 2.5: Comparison of final LLM performance/LLM-annotator agreement broken down by **version** (column blocks) and **feature** (rows). P=Precision, R=Recall, F_1 = F_1 -score, κ =Cohen’s κ , α =Krippendorff’s α .

2.6.3 Manual checks

In addition to the comparison metrics described above, targeted manual checks were conducted in order to better interpret the types of errors that continue to persist. The results revealed that not only is frequency of occurrence a significant factor in the apparent low F_1 scores of low-frequency features (e.g., AktAct, ASUnrg, and DtUniv), but also that a substantial proportion of apparent errors were in fact represent either (i) cases of feature ambiguity, where multiple analyses are defensible, or (ii) human inconsistency or error, where the model output was equally or more internally consistent. For example, the total disagreement rates (calculated as proportion of disagreements to total judgments) for some of the lower F_1 features included 5.5% for ASUnrg, 1.7% for AktAct, and 1.2% for TLTS.

Ambiguity is particularly common in the case of features that express subtle distinctions of nuance that can be difficult to identify reliably in any text, which are in turn exacerbated by the often archaic, poetic, and/or vague language found in English translations of the New Testament Bible. For example, consider the text in (5), from Matthew 18:31.

- (5) “When the other servants saw what happened, they were very sad. They told their lord what had been done.”

Feature(s)	Text	Target Gram	Annotation	Type
DtExp/DtRes	“The man said to him: I have done all these things”	[‘have done’]	1/1	ambiguity
DtRes	‘You should have put my money in the bank’	[‘should have put’]	1	human error
AktAcc	‘Jesus said to them: “Have you read the Scriptures’ ”	[have read]	FN	interpretive error

Table 2.6: Examples of three types of disagreements with human annotations

Presented with this context and tasked with determining the feature profile of the gram [had been done], defensible arguments can be made for both achievement and accomplishment Aktionsarten, as well as experiential and resultative readings. As these and similar annotation challenges occur frequently in the source text, it has been determined that for the purposes of OFM, in such cases the framework is best served by annotating both possible features as present, though this decision had not yet been reached when the original annotations were conducted, which also contributes to misleadingly low F_1 scores for some features.

It should also be noted that among the PLG reading features a major contributing factor to disagreement rates was a shift in annotation strategies which occurred after the initial human annotations were completed but before the final set of experiments were run; namely, it was decided that tokens that occur under scope of irrealis mood would be annotated “0” for all PLG readings. Because the original annotations differ in this regard, a significant number of human-model disagreements among these annotations owe to this factor, representing approximately 35% of all DtExp disagreements, approximately 20% of all DtRes disagreements, and slightly over 50% of all DtUniv disagreements.

Once manual checks had been completed, it became clear that the model’s annotations differ systematically in their accuracy depending on the degree to which they rely on surface forms versus semantic interpretations, motivating a differentiated approach to verification that directly informs the implementation strategy adopted in this project.

2.6.4 Persistent disagreements

Persisting disagreements between human and model annotations fall into three broad categories: genuine semantic ambiguity, inconsistency in human annotations, and model reasoning misinterpretations.

Genuine semantic ambiguities primarily arise in the PLG reading (DtExp, DtRes, DtUniv) and Aktionsart (AktAcc, AktAch, AktAct, AktState) features which are most affected by subtleties of nuance. For example, in the phrase, “no one has fed the cat”, even absent the basic verbal negator “not” the clause still unambiguously conveys the non-occurrence of an event of feeding the cat. By contrast, in the phrase, “he has told us these things”, well-reasoned arguments can be made for both experiential (i.e., focusing on an event in the past history of the participants) and resultative (focusing on the state in which the listeners are now aware of ‘these things’) readings. In such cases, this system seeks to strike a balance between capturing such genuine ambiguities by way of positive

annotation of both features, though this risks incurring an abundance of false positive annotations and such balance always carries some degree of risk in either or both directions. In the case of CRI annotations, this is addressed in consistently high-performing features (e.g., AktAch, TLAspPfv) through verifications which flag likely candidates for false positives as well as negatives, in through manually verifying values for less high-performing features (e.g., AktAct, DtUniv).

Inconsistency in human annotations reflects a variety of potential causes, including insufficiently explicit feature diagnostics, “drift” effects wherein a human’s interpretation of feature diagnostics may drift over the process of annotating large numbers of tokens, and simple errors arising from careless misreading of text or momentary inattentiveness. This is an area in which the model’s consistency of reasoning has a “stabilizing” effect, such that as its reasoning is highly constrained by the narrow set of diagnostics it is provided with and its attentiveness does not waver from token to token, this pushes the human annotation “editor” to confront potential inconsistencies in their own reasoning.

Disagreements owing to the model’s interpretations of the text are particularly challenging to surmount. For example, if the model confuses the focus of a clause as being on a change of state event rather than on the state which results from that event, this requires teaching the model to “understand” the concept of focus and how to evaluate it across any linguistic context it might encounter. Similarly, if the model interprets a punctual event as having a durative process phase, or vice versa, addressing this requires not only robust definition of the concept itself but also teaching the model how to distinguish between, say, “come” in the sense of a durative directed motion event “come” in the sense of a punctual event of arrival. In Table 2.6, the model gives the following reasoning:

- (6) “The verb ‘have read’ is construed as a punctual transition into a state of awareness or understanding (whether the addressee has encountered the Scriptures), lacking an encoded durative process phase or defined goal that culminates in a result state.”

The example given in (6) illustrates a context that is unlikely to be taken into account in the design process of a CRI system, as such a system needs to be able to accommodate texts of virtually any genre, and so the model’s erroneous interpretation of “read the Scriptures” as a punctual state change event cannot be predicted or planned for without taking every single possible linguistic context into consideration. What this brings into focus is the true scope of what it means for a model to actually interpret and generate human-like communication, rather than simply predict a string of human-sounding words.

2.7 Implementation

This system is used to aid in the production of high-quality feature annotations across the datasets that will be used in conducting OFM experiments in the subsequent project. Given the

varied results across features, different degrees of human verification are motivated depending the feature in question.

- (7) a. Type I: features that perform very reliably and require minimal verification: ASTrns, ASUncc, ClCntfcl, ClNegPol, TLMdIrr, TLTS, VcPass
- b. Type II: features that perform broadly reliably and require only targeted human review: AktAch, ASUnrg, TLAspPfv
- c. Type III: highly nuanced features that require careful interpretation and thus more in-depth human review: AktAcc, AktAch, AktState, DtExp, DtRes, DtUniv

The CRI system has been shown to produce reliable annotations for features that depend primarily on surface cues, such as word order, word choice, and morphological marking. All feature types will be reviewed by a human annotator to ensure that annotations are reliable, though different degrees of human supervision are motivated for each type. Type I and II features will make use of the verification system presented in §2.5.3, which flags tokens with rare or otherwise unexpected feature values, such as multiple PLG readings or Aktionsart, lack of argument structure valuation, and uncommon pairings with perfective aspect. In addition to this, Type II features will also be subjected to randomized token checks to verify overall reliability of performance. Type III features present the greatest challenge, and as such will be considered provisional until verified by a human annotator in order to ensure accuracy. In the case of tokens which exhibit genuine ambiguity and for which multiple defensible feature valuations can be made, all possible features will be marked positive in order to capture the greatest breadth of semantic distributions of those features.